

Experiment 9

Student Name: Nikhil Kumar

UID: 25MCI10036

Branch: MCA (AI/ML)

Section/Group: 25MAM-1(A)

Semester: 2-SEM

Date of Performance: 07/04/2026

Subject Name: Technical Training - 1

Subject Code: 25CAP-652

1. Aim of the Session:

To implement database triggers in PostgreSQL to automatically calculate values and enforce constraints during data insertion operations.

2. Software Requirements:

- PostgreSQL
- pgAdmin
- Windows Operating System

3. Objectives of the Session:

- **Trigger Implementation:** To understand how to create and use triggers in PostgreSQL.
- **Automation:** To automate calculation of total payable amount using triggers.
- **Data Integrity:** To enforce constraints using trigger conditions.
- **Real-time Processing:** To execute logic automatically before inserting data.

4. Theory:

Triggers are special database objects that automatically execute a function when a specified event (INSERT, UPDATE, DELETE) occurs on a table. In PostgreSQL, triggers are associated with trigger functions written in PL/pgSQL. They help maintain data integrity, enforce business rules, and automate tasks.

5. Practical / Experiment Steps:

Step 0: Create table query

Code:

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    working_hours INT,  
    perhour_salary NUMERIC,  
    total_payable_amount NUMERIC  
);
```

Step 1: Creating trigger function query.

Code:

```
CREATE OR REPLACE FUNCTION CACULATE_PAYABLE_AMOUNT() RETURNS  
TRIGGER  
AS  
$$  
BEGIN  
    NEW.total_payable_amount:=NEW.perhour_salary*New.working_hours;  
  
    IF NEW.total_payable_amount>25000 THEN  
        RAISE EXCEPTION 'INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN  
NOT BE GREATER THAN 25000';  
    END IF;  
  
    RETURN NEW;  
END;  
  
$$ LANGUAGE PLPGSQL;
```

Step 2: Creating trigger query.

Code:

```
CREATE OR REPLACE TRIGGER AUTOMATED_PAYABLE_AMOUNT_CALCULATION  
BEFORE INSERT  
ON employee  
FOR EACH ROW  
EXECUTE FUNCTION CACULATE_PAYABLE_AMOUNT()
```

Step 3: Inserting valid data.

Code:

```
INSERT INTO EMPLOYEE (EMP_ID, EMP_NAME, working_hours, perhour_salary)  
VALUES (1, 'AKASH', 10, 1000);
```

Output:

Data Output	Messages	Notifications
INSERT 0 1		
Query returned successfully in 1 secs 45 msec.		

Step 4: Inserting Invalid data (Exception case).

Code:

```
INSERT INTO EMPLOYEE (EMP_ID, EMP_NAME, working_hours, perhour_salary)  
VALUES (2, 'ANKUSH', 8, 100000);
```

Output:

Data Output	Messages	Notifications
ERROR: INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN NOT BE GREATER THAN 25000 CONTEXT: PL/pgSQL function caculate_payable_amount() line 6 at RAISE		
SQL state: P0001		

Step 5: Retrieving data query.

Code:

```
SELECT * FROM EMPLOYEE;
```

Output:

Data Output Messages Notifications					
					
			SQL	Showing rows: 1 to 1 	
	emp_id [PK] integer 	emp_name character varying (50) 	working_hours integer 	perhour_salary numeric 	total_payable_amount numeric 
1	1	AKASH	10	1000	10000

6. Learning Outcomes (What I have learned):

- Students will be able to create and use triggers in PostgreSQL.
- Students will understand how to automate calculations using triggers.
- Students will enforce constraints using trigger conditions.
- Students will understand real-time execution of database logic.